

Physical activity increases the bioavailability of flavanones after dietary aronia-citrus juice intake in triathletes.

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Control and triathlete volunteers (n=8 and n=15, respectively) were given 400 mL and 200 mL of aronia-citrus juice (AC-juice), respectively. The 24h urine samples were hydrolysed to determine the flavanones concentration by UPLC-QqQ-MS/MS. The flavanones metabolites in both groups of volunteers were glucuronides, sulfates, and sulfo-glucuronides, and the total excretion of flavanones increased fivefold in the triathletes compared with the control volunteers. The increase of ninefold in the homoeriodictyol of triathletes compared to control volunteers may suggest the overactivation of the microbiota metabolism caused by physical exercise. No differences concerning the bioavailability were detected between men and women in controlboth groups. The AC-juice could provide synergistic effects on health due to the increase in the bioavailability of flavanones, avoiding the deleterious effects caused by the overdosage of nutritional supplements. Copyright © 2012 Elsevier Ltd. All rights reserved.